

Program 4:

Integrate Git with Docker for Source-Controlled Application Builds

Project Structure:

program_4/

—>Dockerfile

—>index.js

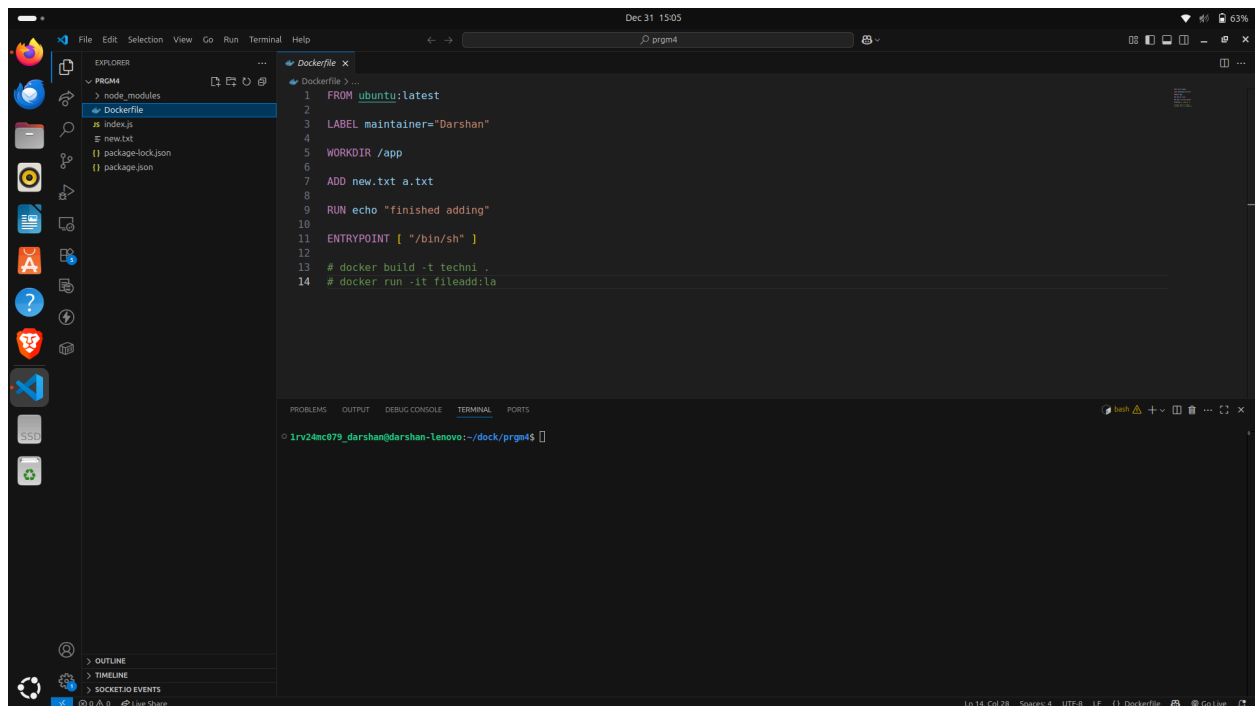
—>package.json

—>package-lock.json

Procedure:

Create a folder (ex: program_4)

Step 1: Create a Dockerfile without any extensions and add the following content

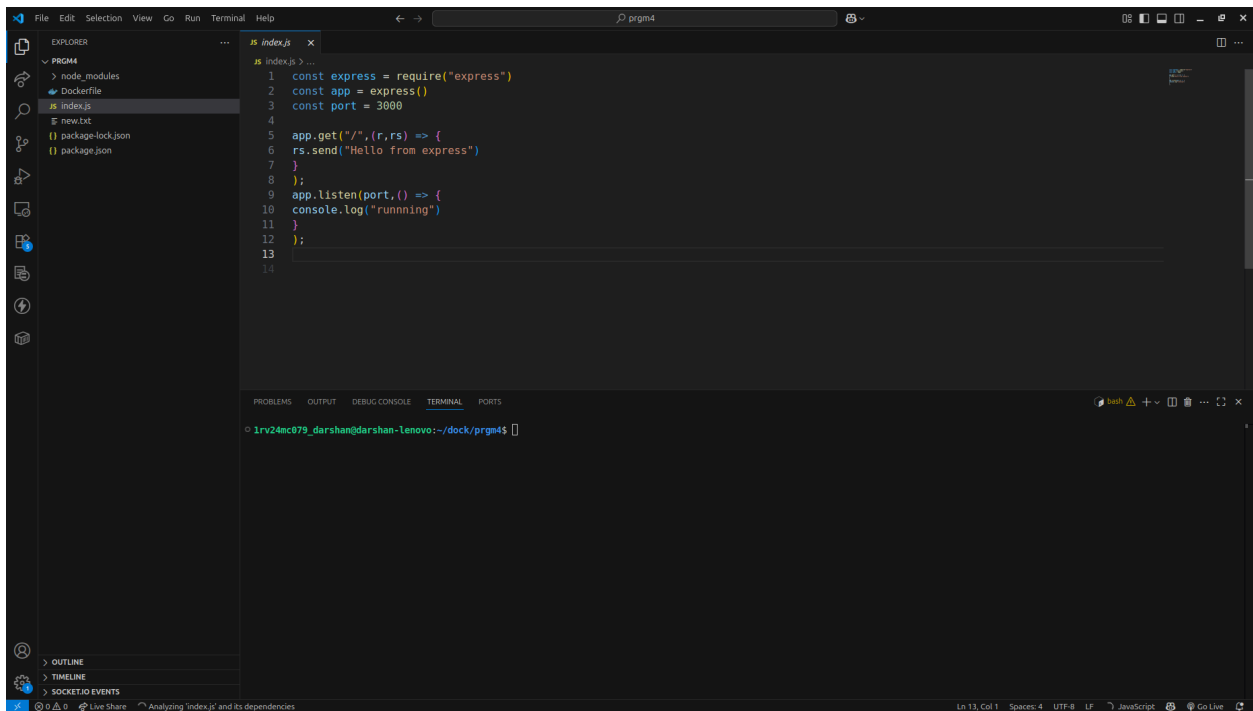


The screenshot shows the Visual Studio Code editor interface. The Explorer panel on the left shows a project named 'PRGM4' with files: 'node_modules', 'Dockerfile', 'index.js', 'new.txt', 'package-lock.json', and 'package.json'. The Dockerfile is open in the editor, showing the following content:

```
1 FROM ubuntu:latest
2
3 LABEL maintainer="Darshan"
4
5 WORKDIR /app
6
7 ADD new.txt a.txt
8
9 RUN echo "finished adding"
10
11 ENTRYPOINT [ "/bin/sh" ]
12
13 # docker build -t techn1 .
14 # docker run -it fileadd:la
```

The Terminal panel at the bottom shows a shell prompt: `1rv24mc979_darshan@darshan-Lenovo:~/dock/prgm4$`. The status bar at the bottom right indicates 'Ln 14, Col 28', 'Spaces: 4', 'UTF-8', 'LF', and 'Dockerfile'.

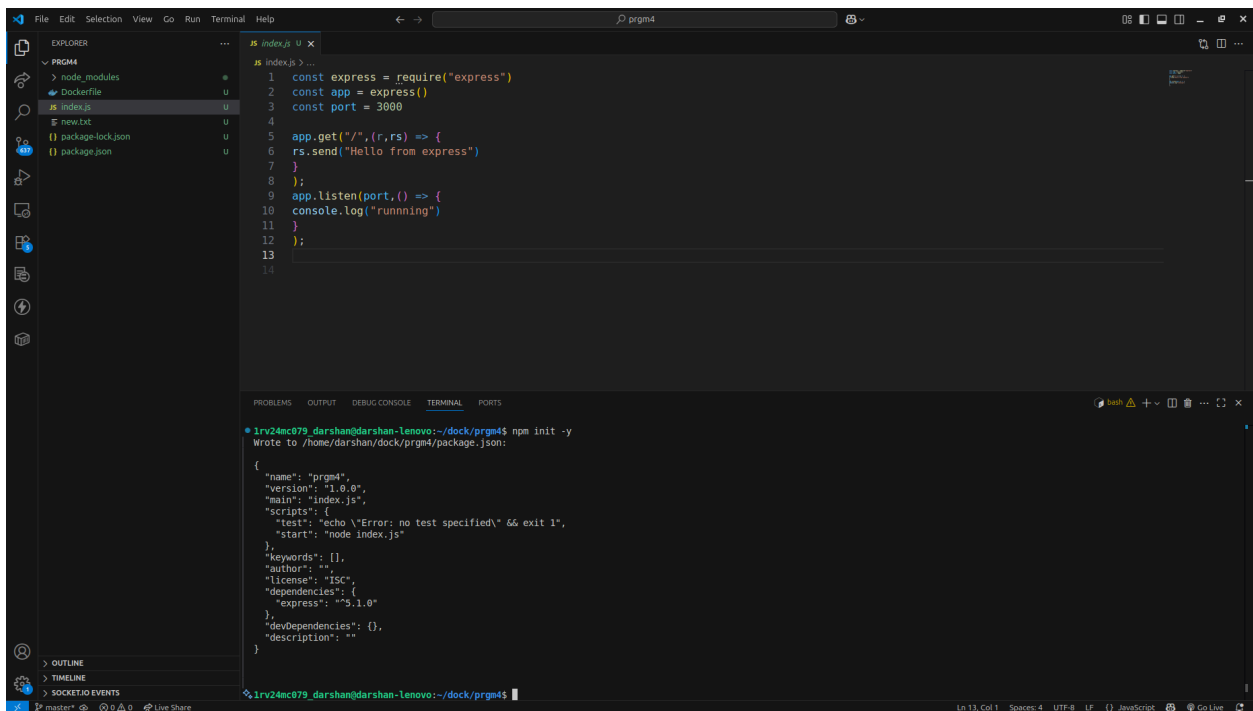
Step 2: Create an index.js file with following code



The screenshot shows the VS Code editor interface. The Explorer sidebar on the left displays the file structure for a project named 'prgm4', including 'node_modules', 'Dockerfile', 'index.js', 'new.txt', 'package-lock.json', and 'package.json'. The main editor area shows the content of 'index.js', which is a simple Express.js server. The terminal at the bottom shows the command prompt '1rv24mc079_darshan@darshan-lenovo:~/dock/prgm4\$'.

```
1 const express = require("express")
2 const app = express()
3 const port = 3000
4
5 app.get("/", (r, rs) => {
6   rs.send("Hello from express")
7 })
8
9 app.listen(port, () => {
10  console.log("running")
11 })
12
13
```

Step 3: create file package.json by executing the command npm init -y



The screenshot shows the VS Code editor interface after running the command 'npm init -y'. The Explorer sidebar on the left now includes 'package-lock.json' and 'package.json' in the file list. The main editor area still shows the content of 'index.js'. The terminal at the bottom shows the command '1rv24mc079_darshan@darshan-lenovo:~/dock/prgm4\$ npm init -y' and the resulting 'package.json' file content.

```
1rv24mc079_darshan@darshan-lenovo:~/dock/prgm4$ npm init -y
Wrote to /home/darshan/dock/prgm4/package.json:

{
  "name": "prgm4",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node index.js"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^5.1.0"
  },
  "devDependencies": {},
  "description": ""
}
```

Step 4: Initialize Git Repository locally.

```
git config --global user.name " "
```

```
git config --global user.email " "
```

```
git init
```

```
git add .
```

```
git commit -m "Initial commit"
```

Step 5: Link to remote repository

Step 6: Push code to GitHub

```
git push -u origin main
```